RAGHAV RAMAN

Data Engineer | ETL & Data Integration | AI Engineer | Informatica IICS, SQL, Python

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Portfolio: theraghavraman.github.io/the_raghav_raman

SCAN FOR PORTFOLIO

PROFESSIONAL SUMMARY

Senior Software Engineer at Capgemini with 4+ years in data engineering: ETL pipelines on Informatica IICS and PowerCenter, Snowflake and SQL, turning raw data into decisions. Outside work I build and ship software: ReData, an incremental data-ingestion engine on DuckDB; Mind Games, a 54-game Linux desktop app on the Snap Store with CI that tests every release under real snapd; and Omni Suite, a privacy-first, offline file toolkit running in the browser with WebAssembly. Now going deeper into AI engineering: AMD ROCm Certified Associate, DeepLearning.AI Deep Learning Specialization, and Knowledge Graphs for RAG.

TECHNICAL SKILLS

ETL & Data Integration: Informatica IICS/IDMC, PowerCenter, Informatica Admin, SSIS, Talend, ETL Pipelines, SCD Type 1/2, Hierarchy Parser, Data Mapping, Data Validation, Flat-File Ingestion, Incremental Sync

Databases & SQL: SQL, Oracle, IBM DB2, SQL Server (SSMS), Snowflake, MySQL, DuckDB, MongoDB

Programming & APIs: Python, FastAPI, REST APIs, SQLAlchemy, Pydantic, Pytest, Celery, Redis, JavaScript, TypeScript

Cloud & Big Data: Azure Synapse, Azure VDI, AWS, GCP, BigQuery, Databricks, Apache Kafka, Apache Arrow, Docker

AI & Retrieval: RAG, LangChain, FAISS, Embeddings, OpenAI API, AI Agents, Prompt Engineering

Data Reliability & Ops: Pipeline Monitoring, ServiceNow, Incident Mgmt, ITIL, Root-Cause Analysis

Tools & Platforms: Git, GitHub, GitHub Actions, Linux/Unix, Nginx, Windows, macOS, Microsoft Excel, BI Reporting

PROFESSIONAL EXPERIENCE

Senior Software Engineer, Data & AI | Capgemini

Oct 2026 – Present

- Collaborated with the team on bridge calls for Informatica updates, coordinating changes and resolving issues live.
- Supported a database migration with the team, moving and verifying ETL jobs and data on the new database.

Analyst, Data Engineering | Capgemini

Oct 2022 – Oct 2026

- Built and maintained ETL pipelines in Informatica PowerCenter and Informatica Intelligent Cloud Services (IICS) for client data platforms.
- Loaded and managed data in Snowflake, and automated routine jobs with shell scripts.
- Built Informatica Cloud (IICS) mappings and tasks for SCD Type 1, SCD Type 2, file handling and Hierarchy Parser use cases on Oracle and flat-file connections.
- Used Router, Aggregator, Normalizer, Joiner, Lookup, Expression, Update Strategy and Sequence Generator transformations; validated loads with row counts and rejected-row checks.
- Worked with business and client teams to turn requirements into reliable data flows, and prepared clean, validated datasets for BI reporting and analytics.
- Reviewed use cases from technical and functional perspectives with teammates; shortlisted for a BFSI data-integration engagement.

PROJECTS

ReData: Incremental Data-Ingestion Engine

Python | DuckDB | Apache Arrow | FastAPI | CLI

Repo: github.com/theraghavraman/redata | Live demo: theraghavraman.github.io/redata

- Built an engine that copies database tables, files and API responses into DuckDB and keeps them current, with connectors for PostgreSQL, MySQL, SQL Server, Snowflake, SQLite, S3-compatible buckets and JSON REST APIs, read as Arrow batches of 50,000 rows.
- Implemented three sync modes (full refresh that builds beside the old table and swaps it in, append by cursor, upsert by key in one transaction), saving cursor state after every successful run and handling schema drift by adding columns and widening types.

- Exposed it through a CLI, a FastAPI HTTP API and a scheduler with a run log queryable in SQL; `redata sync --all` exits with status 1 if any sync fails, so it can run unattended from cron.

ETL Validation Toolkit Python Pydantic Pytest MySQL

Repo: github.com/theraghavraman/etl-validation-toolkit

- Created a reusable validation layer that cleans CSV, Excel and API feeds before MySQL loading, with Pydantic schema checks, typed models, rejected-row reporting and automated tests.

CodeProof: Browser-First Code Verification Lab JavaScript TypeScript Web Workers Pyodide GitHub Actions

Repo: github.com/theraghavraman/codeproof | Live site: theraghavraman.github.io/codeproof

- Built a tool that inspects, type-checks and test-runs unfamiliar JavaScript, TypeScript and Python entirely in the browser tab, so code is never uploaded to a server.
- Ran static rules, a TypeScript type check and import-graph mapping in a Web Worker, executed tests only inside a sandboxed iframe, and ran Python through a pinned Pyodide runtime; findings export as JSON and Markdown evidence reports.
- Set up CI that rebuilds the bundled TypeScript compiler from verified parts and checks every hash before deploying.

RAG Document Q&A API Python FastAPI LangChain FAISS Redis Celery

- Built a document question-answering API covering ingestion, chunking, embeddings, FAISS vector indexing and grounded retrieval.
- Moved ingestion to Celery/Redis background jobs and returned structured JSON answers through a REST endpoint.

AI Code Testing & Bug Investigation Agent Python FastAPI OpenAI API Pytest Pydantic Docker

- Built a tool-calling AI agent that reads source and specification files, generates targeted Pytest cases and runs them to investigate bugs.
- Sandboxed generated tests in Docker with resource and time limits and no network access; produced JSON/Markdown reports separating test evidence from model analysis.

Mind Games Electron Snapcraft GitHub Actions JavaScript

Repo: github.com/theraghavraman/mindgames | Snap Store: snapcraft.io/mind-games

- Published a 54-game offline puzzle app as a strictly confined Linux snap on the Snap Store (`snap install mind-games`), built with Electron under the MIT licence.
- Set up a GitHub Actions pipeline that builds each release and tests it under real snapd, including a nested GNOME Wayland session that launches all 54 games.

Omni Suite JavaScript HTML CSS

Repo: github.com/theraghavraman/omni_suite_light

- Shipped a live, local-first browser suite for PDF and file workflows (merge, split, convert, OCR, compress) with no server-side file handling.
- Selected for the XERJ Corpus Hub, an open code library AI coding agents search for worked examples.

EDUCATION

Bachelor of Technology (B.Tech), Electronics and Communication Engineering

Lakshmi Narain College of Technology (LNCT), Bhopal – RGPV | 2022 | CGPA 7.89/10

CERTIFICATIONS

- Oracle Cloud Infrastructure 2024 Generative AI Certified Professional – Oracle
- IDMC Foundation Certification; Cloud Data Integration for PowerCenter Developers (Foundation) – Informatica
- Generative AI Fundamentals Accreditation – Databricks Academy
- Google Advanced Data Analytics Professional Certificate – Google
- AWS Cloud Quest: Cloud Practitioner; Generative AI Essentials – Amazon Web Services
- Snowflake Data Warehousing Workshop – Snowflake
- Confluent Fundamentals: Apache Kafka – Confluent

ADDITIONAL HIGHLIGHTS

- Completed about 40 hands-on labs in Google Cloud Computing Foundations and 16 labs in the Google Cloud Hero challenge; finished AWS Cloud Quest with 12 DIY assignments.
- Completed the LevelUp SkillUp business-communication program (Mar – Jul 2024).